

PINN - Transonic Flow in a De Laval Nozzle

Sören Greb

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1 Introduction

The present document serves as supplementary material for a prototypical Physics Informed Neural Network (PINN) implemented in PyTorch [Paszke et al., 2017]. The PINN represents the transonic flow in a De Laval nozzle with variable geometry and can be evaluated via an interactive web-based visualization dashboard (see Fig. 1). In Sec. 2, the governing equations and boundary conditions are presented. The network architecture and the multi-objective loss-function are described in Sec. 3. A test case and results are presented in Sec. 4.

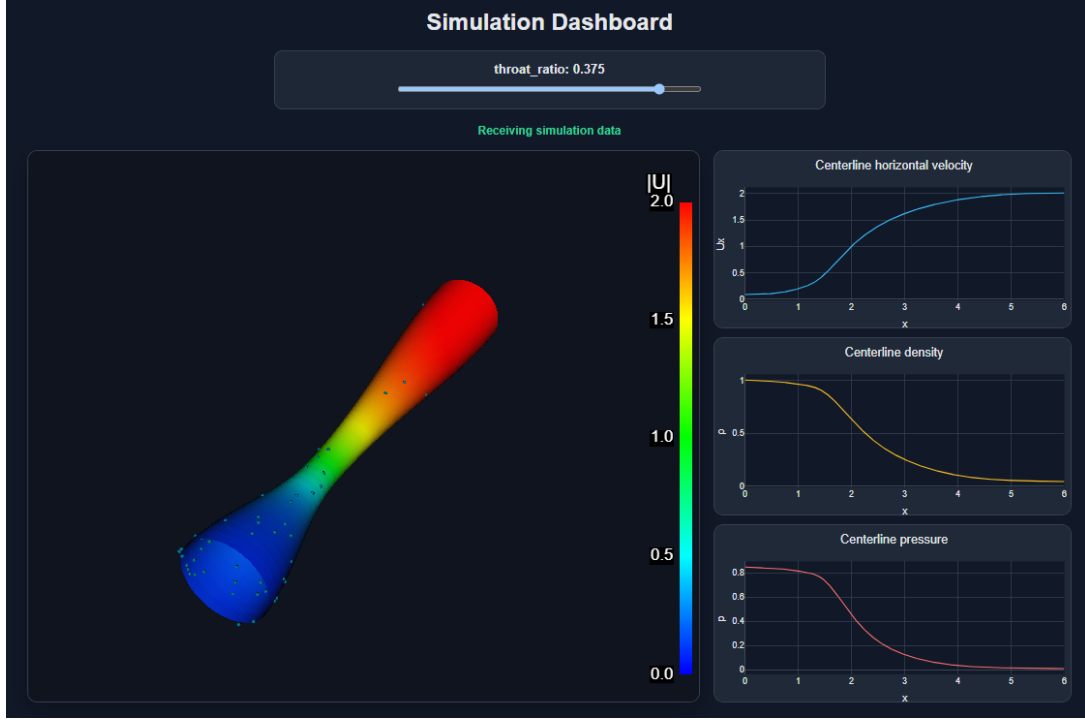


Figure 1: Interactive web-based dashboard for realtime visualization. In the 3D-scene, the geometry can be rotated and the geometry can be manipulated via the slider. The flow direction and magnitude is visualized by Lagrangian particle tracking. The graphs on the right depict the distributions of horizontal flow velocity, density, and pressure, along the the nozzle centerline.

2 Physical Set-Up

2.1 Definitions

- The Einstein notation is used, implying summation over repeated indices, whereas the free indices indicate the tensorial rank of an expression.
- The notation $x|_j$ specifies the j -th component of the vector x_j (point-wise evaluation).
- A set of K indices with upper bounds N_k is defined as $i_k \in \{1, \dots, N_k\}$, where $k = 1, \dots, K$.
- The symbol $\delta_{\eta\nu}$ is the Kronecker delta, where $\delta_{\eta\nu} = 1$ for $\eta = \nu$ and $\delta_{\eta\nu} = 0$ for $\eta \neq \nu$.

2.2 Governing Equations

The PINN predicts the primitive variables density, flow velocity, and pressure,

$$(\rho, u_1, u_2, u_3, p), \quad \text{where } u_\nu = (u_1, u_2, u_3), \quad (1)$$

in a 3-dimensional Cartesian space with coordinates

$$(x_1, x_2, x_3). \quad (2)$$

The governing equations for the conservation of mass, momentum and energy are given by the Euler equations for a steady-state compressible flow:

$$\frac{\partial}{\partial x_\nu} (\rho u_\nu) = 0 \quad (3)$$

$$\frac{\partial}{\partial x_\nu} (\rho u_\eta u_\nu + p \delta_{\eta\nu}) = 0 \quad (4)$$

$$\frac{\partial}{\partial x_\nu} ((E + p) u_\nu) = 0 \quad (5)$$

The total Energy for a calorically perfect gas is given by

$$E = \frac{1}{2} \rho u_\nu u_\nu + \frac{p}{\gamma - 1}. \quad (6)$$

The Mach number of the flow is given by

$$M = \frac{\sqrt{u_\nu u_\nu}}{c}. \quad (7)$$

With the adiabatic index $\gamma = \text{const}$, the isentropic speed of sound,

$$c = \sqrt{\frac{\gamma p}{\rho}}, \quad (8)$$

is governed by the local pressure to density ratio.

2.3 Sonic Transition

The favorable transonic conditions can be determined by assuming an approximately axisymmetric flow with variable cross-sectional area $A(x_1)$, so that Eqns. (3) and (4) simplify to:

$$\frac{\partial}{\partial x_1} (\rho u_1 A) = 0 \quad (9)$$

$$\frac{\partial}{\partial x_1} (\rho u_1^2 + p) = 0 \quad (10)$$

Combining Eqns. (9) and (10) with the barotropic relation $p = p(\rho) \Rightarrow \partial p / \partial x_1 = c^2 \partial \rho / \partial x_1$, the horizontal velocity gradient is obtained:

$$\frac{\partial u_1}{\partial x_1} = \frac{u_1^2 + c^2}{u_1^2 - c^2} \frac{u_1}{A} \frac{\partial A}{\partial x_1} \quad (11)$$

Eqn. (11) shows (i) that sonic transition can only occur at the nozzle throat, where both $u_1^2 - c^2$ and $\partial A / \partial x_1$ are subject to a sign change, and (ii) that the sonic transition can indeed be shock-free as not only $u_1^2 - c^2$ but also $\partial A / \partial x_1$ tends to zero at the throat. The isentropic mass flow relation is given by [Bar-Meir, 2026]

$$M^* = \frac{A_{\text{throat}}}{A^*} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} (M^*)^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}. \quad (12)$$

Eqn. (12) can be used to (i) determine the required inflow Mach number $M^* = M_{\text{in}}$ at the inlet with area $A^* = A_{\text{in}}$ to obtain a choked nozzle, and (ii) to determine the required outlet area

$A^* = A_{\text{out}}$ to obtain a specific target exit Mach number $M^* = M_{\text{out}}$. The stagnation quantities, associated with a state where the flow is brought to rest isentropically and expressed in terms of (ρ^*, p^*, M^*) , are given by [Bar-Meir, 2026]:

$$\rho_0 = \rho^* \left(1 + \frac{\gamma - 1}{2} (M^*)^2 \right)^{\frac{1}{\gamma - 1}} \quad (13)$$

$$p_0 = p^* \left(1 + \frac{\gamma - 1}{2} (M^*)^2 \right)^{\frac{\gamma}{\gamma - 1}} \quad (14)$$

From Eqn. (14) follows the sonic throat state $(\rho_{\text{throat}}, p_{\text{throat}})$, where $M_{\text{throat}} = 1$. The throat speed of sound follows from Eqn. (8) and Eqn. (7) then yields the horizontal throat speed $u_{1,\text{throat}}$, completing the sonic throat state.

2.4 Nozzle Geometry

The nozzle throat to nozzle inlet radius ratio is defined as

$$\mathcal{G} = \frac{R_{\text{throat}}}{R_{\text{in}}}. \quad (15)$$

The throat ratio $\mathcal{G}$ is varied within a certain range and learned by the PINN. The nozzle shape is described by two cosine functions, defined piece-wise for the upstream (converging) and the downstream (diverging) sections from the nozzle throat, respectively:

$$R(x_1) = \begin{cases} R_{\text{throat}} + \frac{1}{2} (R_{\text{in}} - R_{\text{throat}}) \left[1 + \cos \left(\frac{\pi x_1}{L_{\text{conv}}} \right) \right] & \text{for } 0 \leq x_1 \leq L_{\text{conv}} \\ R_{\text{throat}} + \frac{1}{2} (R_{\text{out}} - R_{\text{throat}}) \left[1 + \cos \left(\frac{\pi (x_1 - L_{\text{conv}})}{L_{\text{div}}} \right) \right] & \text{for } L_{\text{conv}} < x_1 \leq L_{\text{tot}} \end{cases} \quad (16)$$

The piece-wise definition of $R(x_1)$ allows for different lengths of the converging and diverging sections, respectively, where the total nozzle length is $L_{\text{tot}} = L_{\text{conv}} + L_{\text{div}}$. In order to compute the wall normal vector, needed for the slip-wall boundary condition, Eqn. (21), the derivative of $R(x_1)$ is needed:

$$\frac{dR}{dx_1} = \begin{cases} -\frac{2}{2L_{\text{conv}}} (R_{\text{in}} - R_{\text{throat}}) \sin \left(\frac{\pi x_1}{L_{\text{conv}}} \right) & \text{for } 0 \leq x_1 \leq L_{\text{conv}} \\ \frac{2}{2L_{\text{div}}} (R_{\text{out}} - R_{\text{throat}}) \sin \left(\frac{\pi (x_1 - L_{\text{conv}})}{L_{\text{div}}} \right) & \text{for } L_{\text{conv}} < x_1 \leq L_{\text{tot}} \end{cases} \quad (17)$$

The wall normal vector then follows as

$$n_\nu = \frac{(-RdR/dx_1, x_2, x_3)}{|(-RdR/dx_1, x_2, x_3)|}. \quad (18)$$

2.5 Boundary Conditions and Constraints

At the inlet, fixed values are imposed (Dirichlet boundary condition):

$$p = p_{\text{in}}, \quad u_1 = u_{1,\text{in}}(A_{\text{throat}}), \quad u_2 = 0, \quad u_3 = 0, \quad \rho = \rho_{\text{in}} \quad (19)$$

The horizontal inlet speed $u_{1,\text{in}}(A_{\text{throat}})$ is given by the isentropic mass flow relation, Eqn. (12), where Eqn. (12) is solved for the required inlet Mach number $M^* = M_{\text{in}}$ by fixed point iteration and where Eqs. (7) and (8) are used to determine $u_{1,\text{in}}$ based on the identified inlet Mach number and the inlet boundary conditions for density and pressure.

Since all characteristics leave the supersonic outlet, zero gradient (Von Neumann) boundary conditions are imposed at the outlet rather than fixed values:

$$\frac{\partial p}{\partial x_\nu} = 0, \quad \frac{\partial u_1}{\partial x_\nu} = 0, \quad \frac{\partial u_2}{\partial x_\nu} = 0, \quad \frac{\partial u_3}{\partial x_\nu} = 0, \quad \frac{\partial \rho}{\partial x_\nu} = 0 \quad (20)$$

However, using Eqn. (12), the outlet area $A^* = A_{\text{out}}$ is determined in such a way that the exit Mach number becomes $M^* = M_{\text{out}} = \text{const}$ for any throat ratio $\mathcal{G}$. At the solid wall, the slip-wall boundary condition

$$u_\nu n_\nu = 0 \quad (21)$$

is imposed, where n_ν is the wall's normal vector. Finally, the following additional constraints are imposed:

$$\int_{\text{cross sec.}} \rho u_1 dA = \text{const} = \rho u_1 A|_{\text{inlet}} \quad (22)$$

$$(\rho, u_1, p)|_{\text{throat}} = (\rho_{\text{throat}}, u_{1,\text{throat}}, p_{\text{throat}}) \quad (23)$$

Eqn. (22) enforces the mass balance over any nozzle cross-section and Eqn. (23) enforces the sonic state across the throat cross-sectional area. These two constraints are redundant as they are already covered by the governing equations, Eqns. (3), (4), and (5). However, they help in preventing the neural network optimizer from converging towards non-physical configurations of the converged variables $\rho \mathbf{u}$ and $E + p$.

3 PINN Set-Up

This section provides a description of the PINN architecture and describes how the derivatives involved in the governing equations are calculated by automated differentiation as introduced into PyTorch by Paszke et al. [2017]. See [PyTorch, 2026] for the PyTorch autograd documentation.

3.1 Network Architecture and Forward Pass

In this section, X is the input to the neural network, representing a list of collocation points randomly sampled within the computational domain Ω and/or its boundaries, and $\hat{Y}$ is the list of predicted outputs, is specified as:

$$\mathcal{X}_{jnd} \in \mathbb{R}^{J \times N \times D} \mapsto \hat{Y}_{jnq}(X_{jnd}) \in \mathbb{R}^{J \times N \times Q}, \quad \text{where} \quad \begin{cases} j : \text{Spatial collocation point index} \\ n : \text{Throat ratio collocation point index} \\ d : \text{Input coordinate index} \\ q : \text{Primitive variable index} \end{cases} \quad (24)$$

At each collocation point with indices j and n , X_{jnd} is represented by the 4-dimensional vector

$$X_{jnd} = (x_1, x_2, x_3, \mathcal{G})_{jn}, \quad (25)$$

With k indicating the hidden layers, K being the number of hidden layers, $K + 1$ representing the prediction layer, and using Tanh activation, the forward pass involves the following steps,

where point-wise evaluation for each j and n is enabled as per network architecture:

$$\begin{aligned}
X_d|_{jn} &\longrightarrow \left[\tanh \left(W_{i_1 i_0}^{(1)} Z_{i_0}^{(0)} + B_{i_1}^{(1)} \right) = Z_{i_1}^{(1)} \right] \Big|_{jn} \longrightarrow [\dots] \Big|_{jn} \\
&\longrightarrow \left[\tanh \left(W_{i_k i_{k-1}}^{(k)} Z_{i_{k-1}}^{(k-1)} + B_{i_k}^{(k)} \right) = Z_{i_k}^{(k)} \right] \Big|_{jn} \longrightarrow [\dots] \Big|_{jn} \\
&\longrightarrow \left[\tanh \left(W_{i_K i_{K-1}}^{(K)} Z_{i_{K-1}}^{(K-1)} + B_{i_K}^{(K)} \right) = Z_{i_K}^{(K)} \right] \Big|_{jn} \\
&\longrightarrow \left[W_{i_{K+1} i_K}^{(K+1)} Z_{i_K}^{(K)} + B_{i_{K+1}}^{(K+1)} = Z_{i_{K+1}}^{(K+1)} \right] \Big|_{jn} \\
&\xrightarrow{\text{primitive variables}} Z_{i_{K+1}}^{(K+1)} \Big|_{jn} = \hat{Y}_q \Big|_{jn} \xrightarrow{\text{convert to fluxes}} \phi \Big|_{jn}
\end{aligned} \tag{26}$$

The input dimension D must be mapped into the I_1 -dimensional first hidden layer through the first weight matrix $W_{i_1 d}^{(1)}$, whereas the output layer must produce the output dimension Q through the final weight matrix $W_{i_K}^{(K+1)}$:

$$\begin{aligned}
Z_{i_0}^{(0)} &= X_d \in \mathbb{R}^D \\
W_{i_1 i_0}^{(1)} &= W_{i_1 d}^{(1)} \in \mathbb{R}^{I_1 \times D} \\
W_{i_{K+1} i_K}^{(K+1)} &= W_{q i_K}^{(K+1)} \in \mathbb{R}^{Q \times I_K} \\
Z_{i_{K+1}}^{(K+1)} &= \hat{Y}_q \in \mathbb{R}^Q
\end{aligned} \tag{27}$$

3.2 Fluxes and Derivatives

Once the network has predicted the primitive variables $\hat{Y}_q = (\rho, u_1, u_2, u_3, p)$ for all j and n , the conserved variables or Euler fluxes, respectively, can be computed. The fluxes are denoted as ϕ as indicated in the final step of Eqn. (26). In total, the network must compute 15 fluxes and partial derivatives to approximate the solution of the governing equations, Eqns. (3), (4), and (5). The mass fluxes and their corresponding partial derivatives are:

$$\begin{aligned}
\phi_{x_1}^{\text{conti}} &= \rho u_1 \longrightarrow \partial \phi_{x_1}^{\text{conti}} / \partial x_1 = \rho \partial u_1 / \partial x_1 + u_1 \partial \rho / \partial x_1 \\
\phi_{x_2}^{\text{conti}} &= \rho u_2 \longrightarrow \partial \phi_{x_2}^{\text{conti}} / \partial x_2 = \rho \partial u_2 / \partial x_2 + u_2 \partial \rho / \partial x_2 \\
\phi_{x_3}^{\text{conti}} &= \rho u_3 \longrightarrow \partial \phi_{x_3}^{\text{conti}} / \partial x_3 = \rho \partial u_3 / \partial x_3 + u_3 \partial \rho / \partial x_3
\end{aligned} \tag{28}$$

The momentum fluxes and their corresponding partial derivatives are:

$$\begin{aligned}
\phi_{x_1, x_1}^{\text{mom}} &= \rho u_1^2 + p \longrightarrow \partial \phi_{x_1, x_1}^{\text{mom}} / \partial x_1 = u_1^2 \partial \rho / \partial x_1 + 2 \rho u_1 \partial u_1 / \partial x_1 + \partial p / \partial x_1 \\
\phi_{x_1, x_2}^{\text{mom}} &= \rho u_1 u_2 \longrightarrow \partial \phi_{x_1, x_2}^{\text{mom}} / \partial x_1 = u_1 u_2 \partial \rho / \partial x_1 + \rho u_2 \partial u_1 / \partial x_1 + \rho u_1 \partial u_2 / \partial x_1 \\
\phi_{x_1, x_3}^{\text{mom}} &= \rho u_1 u_3 \longrightarrow \partial \phi_{x_1, x_3}^{\text{mom}} / \partial x_1 = u_1 u_3 \partial \rho / \partial x_1 + \rho u_3 \partial u_1 / \partial x_1 + \rho u_1 \partial u_3 / \partial x_1 \\
\phi_{x_2, x_1}^{\text{mom}} &= \rho u_2 u_1 \longrightarrow \partial \phi_{x_2, x_1}^{\text{mom}} / \partial x_2 = u_2 u_1 \partial \rho / \partial x_2 + \rho u_1 \partial u_2 / \partial x_2 + \rho u_2 \partial u_1 / \partial x_2 \\
\phi_{x_2, x_2}^{\text{mom}} &= \rho u_2^2 + p \longrightarrow \partial \phi_{x_2, x_2}^{\text{mom}} / \partial x_2 = u_2^2 \partial \rho / \partial x_2 + 2 \rho u_2 \partial u_2 / \partial x_2 + \partial p / \partial x_2 \\
\phi_{x_2, x_3}^{\text{mom}} &= \rho u_2 u_3 \longrightarrow \partial \phi_{x_2, x_3}^{\text{mom}} / \partial x_2 = u_2 u_3 \partial \rho / \partial x_2 + \rho u_3 \partial u_2 / \partial x_2 + \rho u_2 \partial u_3 / \partial x_2 \\
\phi_{x_3, x_1}^{\text{mom}} &= \rho u_3 u_1 \longrightarrow \partial \phi_{x_3, x_1}^{\text{mom}} / \partial x_3 = u_3 u_1 \partial \rho / \partial x_3 + \rho u_1 \partial u_3 / \partial x_3 + \rho u_3 \partial u_1 / \partial x_3 \\
\phi_{x_3, x_2}^{\text{mom}} &= \rho u_3 u_2 \longrightarrow \partial \phi_{x_3, x_2}^{\text{mom}} / \partial x_3 = u_3 u_2 \partial \rho / \partial x_3 + \rho u_2 \partial u_3 / \partial x_3 + \rho u_3 \partial u_2 / \partial x_3 \\
\phi_{x_3, x_3}^{\text{mom}} &= \rho u_3^2 + p \longrightarrow \partial \phi_{x_3, x_3}^{\text{mom}} / \partial x_3 = u_3^2 \partial \rho / \partial x_3 + 2 \rho u_3 \partial u_3 / \partial x_3 + \partial p / \partial x_3
\end{aligned} \tag{29}$$

The energy fluxes and their corresponding partial derivatives, with E specified by Eqn. (6), are:

$$\begin{aligned}
\phi_{x_1}^{\text{energy}} &= (E + p) u_1 \longrightarrow \partial \phi_{x_1}^{\text{energy}} / \partial x_1 = (u_1^2 / 2) \partial \rho / \partial x_1 + \rho u_1 \partial u_1 / \partial x_1 + (\partial p / \partial x_1) / (\gamma - 1) \\
\phi_{x_2}^{\text{energy}} &= (E + p) u_2 \longrightarrow \partial \phi_{x_2}^{\text{energy}} / \partial x_2 = (u_2^2 / 2) \partial \rho / \partial x_2 + \rho u_2 \partial u_2 / \partial x_2 + (\partial p / \partial x_2) / (\gamma - 1) \\
\phi_{x_3}^{\text{energy}} &= (E + p) u_3 \longrightarrow \partial \phi_{x_3}^{\text{energy}} / \partial x_3 = (u_3^2 / 2) \partial \rho / \partial x_3 + \rho u_3 \partial u_3 / \partial x_3 + (\partial p / \partial x_3) / (\gamma - 1)
\end{aligned} \tag{30}$$

The partial derivatives in Eqns. (28), (29), and (30) are combinations of partial derivatives of the predicted primitive variables with respect to the network inputs and can, therefore, be obtained by automated differentiation. Applying the chain rule to

$$Y = Y \left(Z_{i_{K+1}}^{(K+1)} \left(\dots \left(Z_{i_1}^{(1)} (X_d) \right) \right) \right) \quad (31)$$

yields a non-commutative tensor product. Descending order of multiplication with respect to the index k (backpropagation) is defined as

$$\prod_{k=1}^{\overleftarrow{K}} \Pi_k := \Pi_K \Pi_{K-1} \cdots \Pi_1. \quad (32)$$

With Eqn. (26), the chain rule yields

$$\begin{aligned} \left. \frac{\partial Y}{\partial X_d} \right|_{jn} &= \prod_{k=1}^{\overleftarrow{K+1}} \left\{ \frac{\partial Z_{i_k}^{(k)}}{\partial Z_{i_{k-1}}^{(k-1)}} \right\} \bigg|_{jn} \\ &= W_{i_{K+1}i_K}^{(K+1)} \prod_{k=1}^{\overleftarrow{K}} \left\{ \left[1 - \tanh^2 \left(W_{i_k i_{k-1}}^{(k)} Z_{i_{k-1}}^{(k-1)} + B_{i_k}^{(k)} \right) \right] W_{i_k i_{k-1}}^{(k)} \right\} \bigg|_{jn}. \end{aligned} \quad (33)$$

3.3 Multi-Objective Loss-Function

Several spatial regions (interior, wall, inlet, outlet) constitute sampling points to the multi-objective loss-function. Therefore, the upper bound of the spatial collocation point index j , formerly written as J , now has subscripts indicating the sampling points of the interior domain, where the PDE is solved, the inlet and outlet, the solid wall as well as the sampling regions for the additional constraints. The PDE residual, representing the approximation of Eqns. (3), (4), and (5), is then given by:

$$\begin{aligned} L^{\text{PDE}} &= \frac{1}{N J_{\text{PDE}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{PDE}}} \left[\frac{\partial}{\partial x_\nu} (\rho u_\nu) \right]^2 \bigg|_{jn} \\ &+ \frac{1}{N J_{\text{PDE}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{PDE}}} \left[\frac{\partial}{\partial x_\nu} (\rho u_\eta u_\nu + p \delta_{\eta\nu}) \right]^2 \bigg|_{jn} \\ &+ \frac{1}{N J_{\text{PDE}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{PDE}}} \left[\frac{\partial}{\partial x_\nu} ((E + p) u_\nu) \right]^2 \bigg|_{jn} \end{aligned} \quad (34)$$

The inlet loss, representing the approximation of the inlet boundary condition given by Eqn. (19), is given by:

$$\begin{aligned} L^{\text{in}} &= \frac{1}{N J_{\text{in}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{in}}} (\hat{\rho} - \rho_{\text{in}})^2 \bigg|_{jn} + \frac{1}{N J_{\text{in}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{in}}} (\hat{p} - p_{\text{in}})^2 \bigg|_{jn} \\ &+ \frac{1}{N J_{\text{in}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{in}}} (\hat{u}_1 - u_{1,\text{in}}(A_{\text{throat}}))^2 \bigg|_{jn} \\ &+ \frac{1}{N J_{\text{in}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{in}}} (\hat{u}_2 - 0)^2 \bigg|_{jn} + \frac{1}{N J_{\text{in}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{in}}} (\hat{u}_3 - 0)^2 \bigg|_{jn} \end{aligned} \quad (35)$$

The outlet loss, representing the approximation of the outlet boundary condition given by Eqn. (20), is given by:

$$\begin{aligned}
L^{\text{out}} = & \frac{1}{NJ_{\text{out}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{out}}} \left(\frac{\partial \rho}{\partial x_\nu} \right) \left(\frac{\partial \rho}{\partial x_\nu} \right) \Big|_{jn} + \frac{1}{NJ_{\text{out}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{out}}} \left(\frac{\partial p}{\partial x_\nu} \right) \left(\frac{\partial p}{\partial x_\nu} \right) \Big|_{jn} \\
& + \frac{1}{NJ_{\text{out}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{out}}} \left(\frac{\partial u_1}{\partial x_\nu} \right) \left(\frac{\partial u_1}{\partial x_\nu} \right) \Big|_{jn} \\
& + \frac{1}{NJ_{\text{out}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{out}}} \left(\frac{\partial u_2}{\partial x_\nu} \right) \left(\frac{\partial u_2}{\partial x_\nu} \right) \Big|_{jn} + \frac{1}{NJ_{\text{out}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{out}}} \left(\frac{\partial u_3}{\partial x_\nu} \right) \left(\frac{\partial u_3}{\partial x_\nu} \right) \Big|_{jn}
\end{aligned} \tag{36}$$

The wall loss, representing the approximation of the slip-wall boundary condition Eqn. (21) is given by:

$$L^{\text{wall}} = \frac{1}{NJ_{\text{wall}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{wall}}} (u_\nu n_\nu)^2 \Big|_{jn} \tag{37}$$

The mass loss, representing the approximation of the additional constraint given by Eqn. (22), is given by:

$$L^{\text{mass}} = \frac{1}{NJ_x} \sum_{n=1}^N \sum_{j=1}^{J_x} \left[\rho_{\text{in}} u_{1,\text{in}} A_{\text{in}} - \sum_{i=1}^{I_r, \theta} (\rho u_1 \Delta A)_i \right]^2 \Big|_{jn} \tag{38}$$

The throat loss, representing the approximation of the additional constraint given by Eqn. (23), is given by:

$$\begin{aligned}
L^{\text{throat}} = & \frac{1}{NJ_{\text{throat}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{throat}}} (\hat{\rho} - \rho_{\text{throat}})^2 \Big|_{jn} \\
& + \frac{1}{NJ_{\text{throat}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{throat}}} (|\hat{u}_1| - c_{\text{throat}})^2 \Big|_{jn} \\
& + \frac{1}{NJ_{\text{throat}}} \sum_{n=1}^N \sum_{j=1}^{J_{\text{throat}}} (\hat{p} - p_{\text{throat}})^2 \Big|_{jn}
\end{aligned} \tag{39}$$

The multi-objective loss-function follows as:

$$L = \frac{w^{\text{PDE}} L^{\text{PDE}} + w^{\text{in}} L^{\text{in}} + w^{\text{out}} L^{\text{out}} + w^{\text{wall}} L^{\text{wall}} + w^{\text{mass}} L^{\text{mass}} + w^{\text{throat}} L^{\text{throat}}}{w^{\text{PDE}} + w^{\text{in}} + w^{\text{out}} + w^{\text{wall}} + w^{\text{mass}} + w^{\text{throat}}} \tag{40}$$

For the present configuration as described in Sec. 4.1, $w = 1$ for all weights $w^{\dots}$ in Eq. (40). Strictly speaking, L^{PDE} , L^{in} , and L^{out} are multi-objective loss-functions by themselves. To keep the individual contributions of the loss terms well balanced in terms of magnitude, the physical parameters are considered as dimensionless quantities.

3.4 Network Layout and Optimizer

For the test case presented in 4, the network has been trained with 2 hidden layers and 64 neurons each, where the 4 inputs are the 3 spatial coordinates (see Eqn. (2)) plus the throat ratio (see Eqn. (15)), and where the 5 outputs are the primitive variables given by Eqn. (1). Following the input-output mapping specified by Eqn. (24), the network layout is given by:

$$\underbrace{\text{Tanh}(\text{Linear}(4, 64))}_{\text{Input hidden layer}} \longrightarrow \underbrace{\text{Tanh}(\text{Linear}(64, 64))}_{\text{Hidden layer}} \longrightarrow \underbrace{\text{Linear}(64, 5)}_{\text{Linear output layer}}$$

A stochastic gradient descent optimizer with adaptive moment estimation and weight decay is used (*torch.optim.AdamW*). The weight decay is 1.0e-4. The network is trained for 20000 epochs. Starting with a learning rate of 2.00e-3, the learning rate is reduced by a factor of 0.8 every 1000 epochs, finalizing with a learning rate of 2.88e-5.

4 Test Case

4.1 Test Case Configuration

Tables 1, 2 and 3 show the inlet, throat and outlet conditions following from the boundary conditions and constraints described in Sec. 2.5. The following relations and conditions are established:

- Inlet: $\rho_{\text{in}} = 1.00$, $p_{\text{in}} = 0.85 \rightarrow u_{1,\text{in}}$ and M_{in} depend on the throat ratio $\mathcal{G}$.
- Outlet: $M_{\text{out}} = 3.5 \rightarrow R_{\text{out}}$ depends on $\mathcal{G}$.
- Throat: $M_{\text{throat}} = 1$.
- Total mass flow: Depends on $\mathcal{G}$ as $u_{1,\text{in}}$ depends on $\mathcal{G}$.
- Geometry: $R_{\text{in}} = 0.5$, $L_{\text{conv}} = 2.0$, $L_{\text{div}} = 4.0$.
- Thermodynamic parameters: $\gamma = 1.4 = \text{const} \rightarrow \text{speed of sound } c \propto \sqrt{p/\rho}$.

$R_{\text{throat}}/R_{\text{in}}$	R_{in}	ρ	u_1	p	c	M	$\dot{m}$
0.2	0.500	1.000	0.025	0.850	1.091	0.023	0.020
0.3	0.500	1.000	0.057	0.850	1.091	0.052	0.045
0.4	0.500	1.000	0.102	0.850	1.091	0.093	0.080

Table 1: Inlet conditions for different throat ratios.

$R_{\text{throat}}/R_{\text{in}}$	R_{throat}	ρ	u_1	p	c	M	$\dot{m}$
0.2	0.100	0.634	0.996	0.449	0.996	1.000	0.020
0.3	0.150	0.635	0.996	0.450	0.996	1.000	0.045
0.4	0.200	0.637	0.997	0.452	0.996	1.000	0.080

Table 2: Throat conditions for different throat ratios.

$R_{\text{throat}}/R_{\text{in}}$	R_{out}	ρ	u_1	p	c	M	$\dot{m}$
0.2	0.261	0.045	2.056	0.011	0.587	3.500	0.020
0.3	0.391	0.045	2.056	0.011	0.587	3.500	0.045
0.4	0.521	0.045	2.057	0.011	0.588	3.500	0.080

Table 3: Outlet conditions for different throat ratios.

4.2 Results

Fig. 2 shows the loss histories for the loss-functions given in Sec. 3.3, where the total "loss" is given by Eqn. (40). Both the horizontal and the vertical axes are depicted on logarithmic scales. After a rapid initial decay, the loss decay decelerates over subsequent epochs. After 20000 epochs, all losses have fallen below 1e-3, where the PDE loss exhibits the slowest convergence.

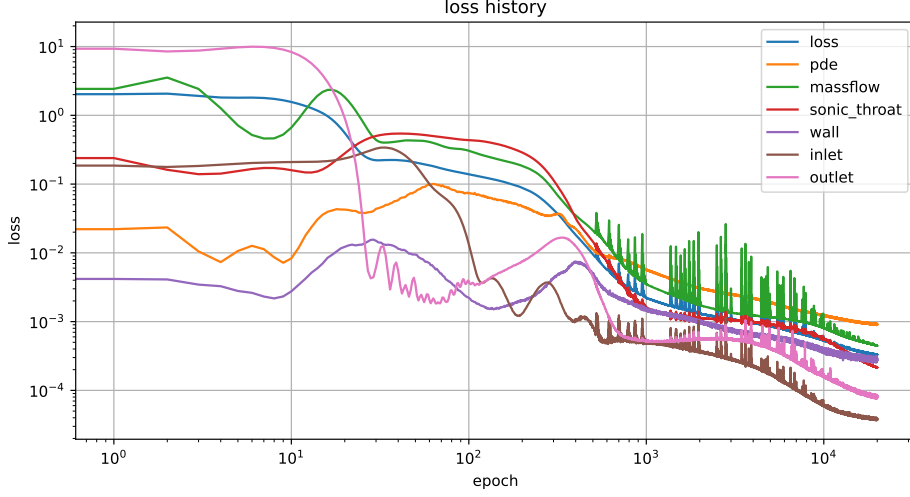


Figure 2: Loss histories for the loss-functions given in Sec. 3.3.

Fig. 3 depicts the horizontal centerline velocity over the epochs, recorded at 7 centerline locations x_1 , denoted in the legend as x , and for 3 different throat ratios $\mathcal{G}$ (see Eqn. (15)), denoted in the legend as "tr". The different centerline locations are distinguished by color, whereas the throat ratios are distinguished by linestyle (solid, dashed, dotted). It can be seen that the flow steadily accelerates towards the outlet for all throat ratios. Tables 1, 2, and 3 suggest that the flow velocities upstream from the throat exhibit a distinct dependency on the throat ratio, which is indeed observed in Fig. 3. At the throat and further downstream, the velocity magnitudes for the different throat ratios should collapse. This is indeed the case, where the throat velocities exhibits the slowest convergence. This is explained by the fact that the throat area exhibits the largest velocity gradient, so that minor deviations from the ideal sonic state result in noticeable discrepancies. A comparison with the reference values given in Tables 1, 2, and 3 shows a satisfactory overall agreement.

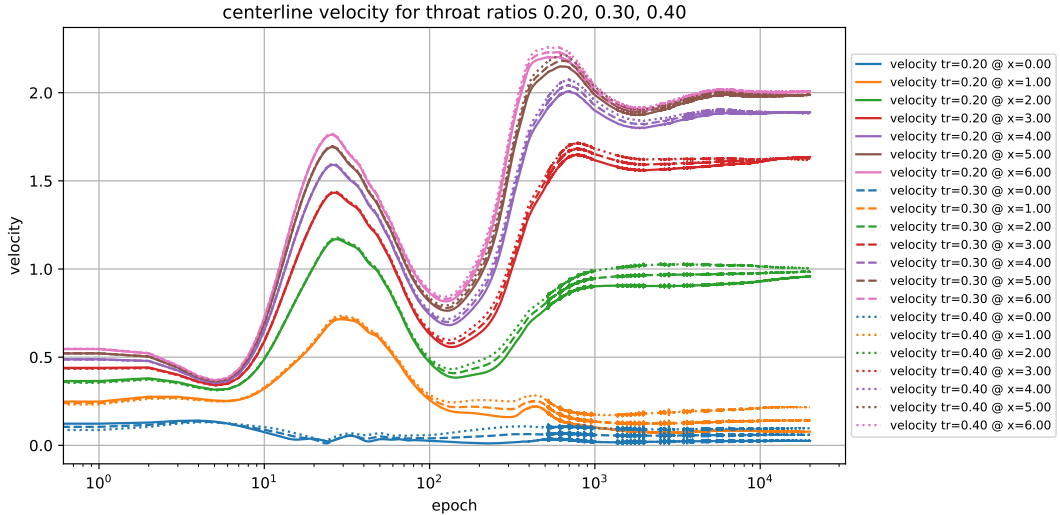


Figure 3: Horizontal centerline velocity over the epochs, recorded at 7 centerline locations x_1 and for 3 different throat ratios $\mathcal{G}$ (see Eqn. (15)), denoted in the legend as "tr".

Fig. 4 depicts the mass flow at 7 centerline locations, obtained by integration over the corresponding cross-sectional area as described by Eqn. (22), for 3 different throat ratios $\mathcal{G}$ (see

Eqn. (15)). Since the inlet density is the same for all throat ratios, whereas the inlet speed depends on the throat ratio, the mass flows differ as per throat ratio but must be conserved along the centerline. This behavior can be observed in Fig. 4. A satisfactory overall agreement with the mass flows for the different throat ratios is achieved. For comparison, see, for instance, Table 1.

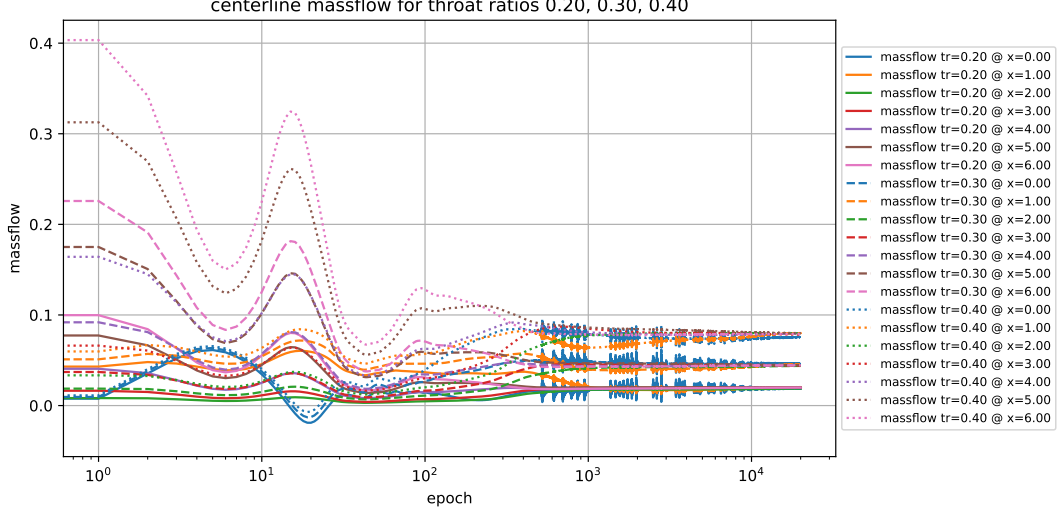


Figure 4: Mass flow at 7 centerline locations, obtained by integration over the corresponding cross-sectional area as described by Eqn. (22), for 3 different throat ratios $\mathcal{G}$ (see Eqn. (15)), denoted in the legend as "tr".

Figs. 5 and 6 depict the density and pressure distributions, respectively, at 7 centerline locations and for 3 different throat ratios $\mathcal{G}$ (see Eqn. (15)). The Dirichlet boundary conditions at the inlet are learned by the PINN. Regarding the throat and outlet, the theoretical values given in Tables 1, 2, and 3 suggest only minor variations of density and pressure over the throat ratio. Figs. 5 and 6 indeed show only minor deviations of density and pressure as per throat ratio at a given centerline location. It is observed that the density at $x = 1.0$, between inlet and throat, requires the largest number of epochs to achieve physical convergence.

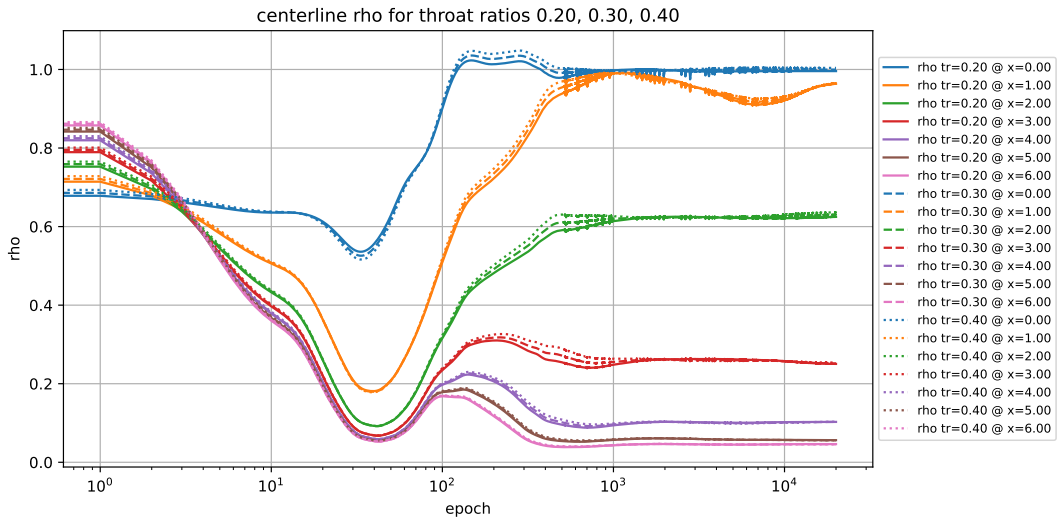


Figure 5: Density at 7 centerline locations and for 3 different throat ratios $\mathcal{G}$ (see Eqn. (15)), denoted in the legend as "tr".

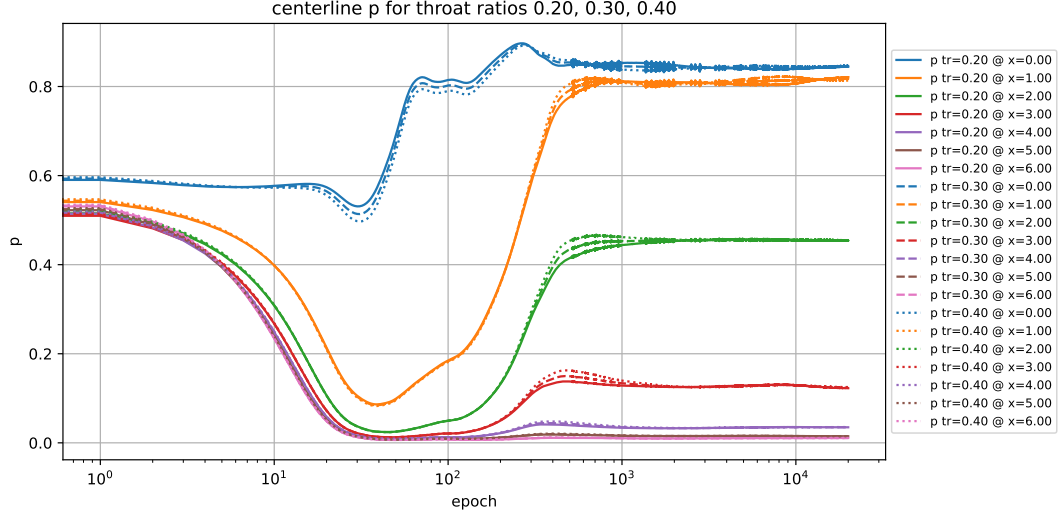


Figure 6: Pressure at 7 centerline locations and for 3 different throat ratios $\mathcal{G}$ (see Eqn. (15)), denoted in the legend as "tr".

Fig. 7 shows longitudinal cross-sectional views of the velocity magnitude distribution for two different throat ratios, $\mathcal{G} = 0.2$ (left) and $\mathcal{G} = 0.4$ (right). It can be seen that while the velocity magnitude exhibits very similar distributions downstream from the throat section, the nozzle with smaller throat ratio exhibits larger velocity gradients in the section upstream from the throat. This is explained by the smaller inlet velocity for $\mathcal{G} = 0.2$ (see Table 1), where the flow is accelerated more rapidly due to the smaller throat cross-sectional area.

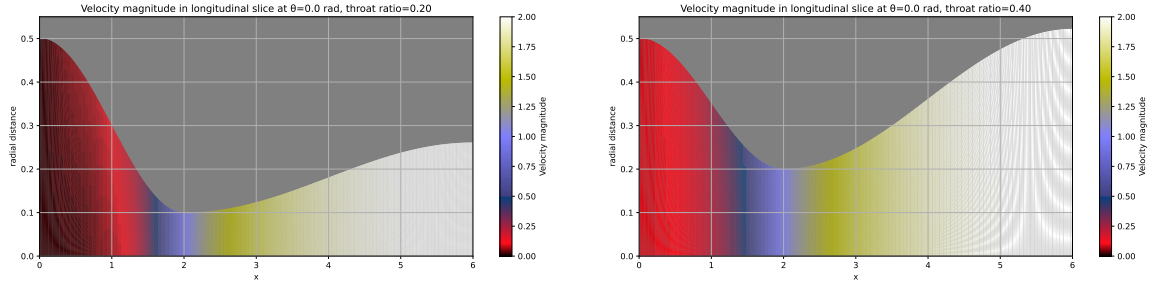


Figure 7: Longitudinal cross-sectional views of the velocity magnitude distribution for two different throat ratios, $\mathcal{G} = 0.2$ (left) and $\mathcal{G} = 0.4$ (right).

5 References

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